

Finite-Prime Stratified Pro-Site Realization

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Summary

Pass 76 builds the first natural model for the Pass-75 support and stage projectors e_p, q_n , replacing the abstract idempotents by geometric operations on a stratified pro-site.

The model is

$$\text{StratPro}_\epsilon(U, N),$$

where $U = \{2, 3, 5, 7, 11, 13\}$ is the checked prime universe carried as a finite discrete (Stone) space and N is the lcm truncation depth. It is still a finite-window certificate model, not yet an LCA-sheaf or condensed realization.

Geometric Projectors

- **Clopen support projectors.** e_p is multiplication by the characteristic function $\mathbf{1}_{\{p\}}$ of the clopen stratum $\{p\} \subseteq U$, so that $e_S = (\cdot) \mathbf{1}_S$ and the Boolean relation becomes pointwise idempotent multiplication:

$$e_S e_T = (\cdot) \mathbf{1}_S \mathbf{1}_T = (\cdot) \mathbf{1}_{S \cap T} = e_{S \cap T}.$$

- **Pro-stage truncation projectors.** q_n is the prefix truncation of the non-Mittag-Leffler lcm tower $K_m = N_m \mathbb{Z}$ at stage n , with $q_n q_m = q_{\min(n, m)}$. The truncations remain cofinal, so the derived datum $\varprojlim^1 K_m \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ survives in the limit.

Site Factorization

The projector realization factors through the site:

$$\rho_{\text{proj}} = (\text{forget site}) \circ \rho_{\text{site}}, \quad \rho_{\text{site}} : \mathcal{H}_\epsilon \rightarrow \text{StratPro}_\epsilon(U, N).$$

Theorem 76b (site factorization and faithfulness). On the checked window, ρ_{site} is faithful on all five generator families: the site signature separates the 75 generators with zero collisions, whereas the plain tag-forgetting signature collapses them to 50.

Theorem 76c (clopen and stage relations). The clopen support projectors realize the finite Boolean algebra of strata (4160 verified $e_S e_T = e_{S \cap T}$ instances); the stage projectors realize the prefix-truncation chain (576 verified $q_n q_m = q_{\min(n, m)}$ instances). The separation is forced exactly at the plain target's collisions, including repeated lcm stages $N_n = N_{n+1}$ (e.g. $N_5 = N_6 = 60$, $N_{13} = N_{14} = N_{15} = 360360$) distinguished only by the stage index n .

Machine Verification

code/scripts/check-pass76.py -> artifacts/reports/pass76-stratified-pro-site-realization-check.json (overall PASS): 75 generators across the five families tested through six primes, $k \leq 3$, and lcm stages through N_{24} ; site and projector global injectivity each with zero collisions while the plain target has 12; site family faithfulness passing on all five families; 4160 clopen Boolean support relations and 576 stage-filtration relations all passing; the factorization of ρ_{proj} certified on the window.

Limit of the Pass

The support and stage projectors now have a genuine geometric reading, but only on a finite discrete prime universe with a truncated tower. The remaining problem is to upgrade $\text{StratPro}_\epsilon(U, N)$ to an all-prime derived exact target -- an LCA sheaf on the profinite prime space, a condensed/solid abelian object, or a canonical exact pro-category -- and to prove the signed duality law $D_{\text{res}}(\epsilon_{\mathbb{P}}) = -\epsilon_{\mathbb{P}}^\vee$ there as a genuine all-prime theorem, or to exhibit the derived/non-Hausdorff barrier that blocks it.